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Solitons in One-Dimensional Molecular Chains

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A theory of collective excitations in one-dimensional chains of weakly bound molecules which correspond to intramolecular excitations is developed. It is shown that these excitations are described by solitary waves (solitons) propagating (without "smearing") along the chain together with local deformation. The theory is developed without applying the assumption of inertialess motion of molecules. It is shown that the ultimate velocity of solitons is the velocity of sound waves corresponding to longitudinal vibrations of molecules. The problem of soliton stability is discussed.

Es wird eine Theorie kollektiver Anregungen in eindimensionalen Ketten von schwach gebundenen Molekülen entwickelt, die intramolekularen Anregungen entsprechen. Es wird gezeigt, daß diese Anregungen beschrieben werden durch solitäre Wellen (Solitonen), die sich (ohne „Verschmierung“) in Richtung der Kette ausbreiten, zusammen mit einer lokalen Deformation. Die Theorie wird ohne die Annahme einer trägheitslosen Bewegung der Moleküle entwickelt. Es wird gezeigt, daß die Grenzgeschwindigkeit der Solitonen die Geschwindigkeit von Schallwellen ist, die den longitudinalen Schwingungen der Moleküle entsprechen. Das Problem der Solitonenstabilität wird diskutiert.

1. Introduction

A special type of collective excitations which has been discussed by the authors in [1] and called particle-like excitations or solitary excitons is possible in one-dimensional molecular chains which consist of periodically repeated molecules (or groups of strongly bound atoms) interacting by relatively weak forces. In what follows these excitations will be referred to as solitons. Particular cases of these systems are α -helical protein molecules in which the peptide groups made up of four atoms (HNCO) bound by chemical forces are arranged along the molecule in the form of three chains. In each chain the neighbouring peptide groups are linked by hydrogen bonds which provide for a rigid structure of α -helix.

The collective excitations of a one-dimensional molecular chain which correspond to intramolecular electronic or vibrational excitations with an electric moment directed along the chain are of a very specific nature different from ordinary excitons in three-dimensional crystals. In protein molecules these excitations include excitations (based on valent Amide I vibrations) with an energy of 0.21 eV and an electric dipole moment of 0.3 Debye directed along the axis of the molecule.

The solitons (solitary excitons) considered in [1] are excitations localized in some region of a chain; they are connected with varying distance between the molecules and propagate with a certain velocity along the chain. The theories of electronic excitations and vibrational amide I excitations of α -helical protein

molecules, developed in [1] and [2], respectively, assumed that the distortion of the chain follows in an inertialess manner the motion of the excitation. This assumption imposes some restrictions on the excitation velocity. In the present paper we develop a theory avoiding the concept of the inertialess motion of the molecules. We show that the ultimate velocity of a soliton in a molecular chain is the velocity of sound waves corresponding to molecular vibrations along the chain. The problem of the soliton stability is also discussed.

2. The Basic Equations and Their Solution in the Form of Solitons

Consider an infinite chain of weakly bound molecules (or groups of atoms) with mass M , which are at distance R . We also assume that the internal excitations of molecules (electronic or vibrational ones) in the chain are characterized by an energy ε and an electric dipole moment d directed along the chain. When the molecule becomes excited the forces of its interaction with neighbouring molecules are changed, this resulting in a change in the equilibrium distances between the molecules. Besides, since the molecules are identical, there arise some additional resonance interactions causing the excitation transfer from one of the molecules to another. The internal excitations of the molecules and their motion about equilibrium positions are inseparably linked.

In a one-dimensional chain it is sufficient to take into account only the interaction between the neighbouring molecules. In this approximation the collective excitations of the chain which correspond to an intramolecular excitation with an energy ε may be described by the Hamiltonian

$$H = T + U + \sum_n [(\varepsilon - D_n) B_n^+ B_n - J(B_{n+1}^+ B_n + B_{n+1} B_n^+)], \quad (1)$$

here the subscript n labels the molecules occupying the positions x_n in the chain; B_n^+ , B_n are the creation and the annihilation operators of an intramolecular excitation. They satisfy the commutation relations

$$[B_n, B_m^+] = \delta_{nm}, \quad [B_n, B_m] = 0.$$

The function $-D_n$ characterizes the change in the static interaction of the n -th molecule with the neighbouring molecules when it is excited. If ϱ_n is defined by the expression

$$x_n - x_{n-1} = R - \varrho_n, \quad (2)$$

then (allowing for the interaction with the nearest neighbours)

$$D_n = \mathcal{D}(|x_{n+1} - x_n|) + \mathcal{D}(|x_{n-1} - x_n|) \approx \left(1 + \frac{\beta}{R} \varrho_n\right) D, \quad (3)$$

where $D \equiv 2\mathcal{D}(R)$ and β are the positive parameters of the theory. The resonance interaction determining the transition of an intramolecular excitation from one molecule to another is characterized by the quantity $-J$ which is defined by the electric dipole moment using the equality

$$J = 2d^2 R^{-3}. \quad (4)$$

The first two terms in (1) correspond to the potential and the kinetic energy of the longitudinal displacements of molecules. The potential energy is chosen in the harmonic approximation:

$$U = \frac{1}{2} w \sum_n \varrho_n^2. \quad (5)$$

When the equation $\dot{x}_n = - \sum_{-\infty < l \leq n} \dot{\varrho}_l$ (5a) is taken into account the kinetic energy can be written as

$$T = \frac{1}{2} M \sum_n \dot{x}_n^2 = \frac{1}{2} M \sum_n \left(\sum_{-\infty < l \leq n} \dot{\varrho}_l \right)^2. \quad (6)$$

Dotted symbols in (6) and in subsequent expressions mean a derivative with respect to time.

The collective excitations of interest are described by the wave function

$$|\psi\rangle = \sum_n a_n(t) B_n^+ |0\rangle. \quad (7)$$

The coefficients $a_n(t)$ satisfy the condition

$$\sum_n |a_n(t)|^2 = 1 \quad (7a)$$

and characterize the distribution of excitation along the chain of molecules. These coefficients satisfy a set of equations which can be obtained from the Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = H |\psi\rangle. \quad (8)$$

Using the explicit form of the operator H and (8), we obtain the equation

$$\sum_n \left\{ i\hbar \frac{\partial a_n}{\partial t} - \left[\varepsilon + T + U - \left(1 + \frac{\beta}{R} \varrho_n \right) D \right] a_n - J(a_{n+1} + a_{n-1}) \right\} B_n^+ |0\rangle = 0.$$

Since the functions $B^+ |0\rangle$ corresponding to different values of n are orthogonal, we can use this equation to obtain the following set of equations:

$$i\hbar \frac{\partial a_n}{\partial t} = \left[\varepsilon + T + U - \left(1 + \frac{\beta}{R} \varrho_n \right) D \right] a_n - J(a_{n+1} + a_{n-1}). \quad (9)$$

It is now convenient to introduce the functional of the variables ϱ_n , $\dot{\varrho}_n$ and a_n

$$\begin{aligned} F &\equiv \langle \psi | H | \psi \rangle = \\ &= \sum_n \left\{ \left[\varepsilon + T + U - \left(1 + \frac{\beta}{R} \varrho_n \right) D \right] a_n^* a_n - J a_n^* (a_{n+1} + a_{n-1}) \right\}. \end{aligned} \quad (10)$$

Following Toda [3], we associate the displacements with canonically conjugated variables

$$s_n = \frac{\partial T}{\partial \dot{\varrho}_n} = -M \sum_{l \geq n} \dot{x}_l.$$

This expression yields the equation

$$\dot{x}_n = \frac{1}{M} (s_n - s_{n-1}),$$

which enables the kinetic energy of the molecular motion to be expressed in terms of the variables

$$T = \frac{1}{2M} \sum_n (s_n - s_{n-1})^2. \quad (11)$$

Thus at fixed values of the coefficients $a_n(t)$ the function F is the Hamiltonian function with respect to the molecular displacements ϱ_n and their

canonical conjugates s_n . We can therefore write the Hamilton equations

$$\dot{q}_n = \frac{\partial F}{\partial s_n} = \frac{1}{M} (2s_n - s_{n+1} - s_{n-1}),$$

$$\dot{s}_n = -\frac{\partial F}{\partial q_n} = -wq_n + \frac{\beta D}{R} |a_n(t)|^2.$$

Eliminating s_n from these equations we find

$$\ddot{q}_n = -\frac{w}{M} (2q_n - q_{n+1} - q_{n-1}) + \frac{\beta D}{MR} (2|a_n|^2 - |a_{n+1}|^2 - |a_{n-1}|^2). \quad (12)$$

The set of equations (9) and (12) defines the collective excitations of a chain. If we are interested only in excitations distributed over some region of a chain, we can apply the continuum model. Let us introduce a dimensionless variable $\xi = x/R$ and continuous functions $q(\xi, t)$ and $a(\xi, t)$ such that

$$q(n, t) = q_n(t), \quad a(n, t) = a_n(t).$$

Using the expressions

$$q(\xi \pm 1, t) \approx q(\xi) \pm \frac{\partial q(\xi, t)}{\partial \xi} + \frac{1}{2} \frac{\partial^2 q(\xi, t)}{\partial \xi^2},$$

$$a(\xi \pm 1, t) \approx a(\xi) \pm \frac{\partial a(\xi, t)}{\partial \xi} + \frac{1}{2} \frac{\partial^2 a(\xi, t)}{\partial \xi^2},$$

$$|a(\xi \pm 1, t)|^2 \approx |a(\xi, t)|^2 \pm \frac{\partial}{\partial \xi} |a(\xi, t)|^2 + \frac{1}{2} \frac{\partial^2}{\partial \xi^2} |a(\xi, t)|^2,$$

we can transform (9) and (12) to

$$i\hbar \frac{\partial a(\xi, t)}{\partial t} - \left[\lambda - \frac{\beta D}{R} q(\xi, t) \right] a(\xi, t) + J \frac{\partial^2 a(\xi, t)}{\partial \xi^2} = 0, \quad (13)$$

$$\frac{\partial^2 q(\xi, t)}{\partial t^2} - v_a^2 \frac{\partial^2 q(\xi, t)}{\partial \xi^2} + \frac{\beta D}{MR} \frac{\partial^2}{\partial \xi^2} |a(\xi, t)|^2 = 0, \quad (14)$$

where

$$\lambda \equiv \varepsilon + T + U - D - 2J, \quad (15)$$

$$T + U = \frac{M}{2} \int_{-\infty}^{\infty} d\xi \left[\left(\int_{-\infty}^{\xi} d\eta \frac{\partial}{\partial t} |q(\eta, t)|^2 \right)^2 + v_a^2 |a(\xi, t)|^4 \right], \quad (16)$$

$v_a = \sqrt{w/R}$, $v_a R = V_a$ is the longitudinal sound velocity in the chain.

We shall look for solutions of the set of (13), (14) in the form of excitations propagating along the chain with the velocity $V = vR$. In other words, we set

$$q(\xi, t) = q(\xi - vt), \quad a(\xi, t) = \Phi(\xi - vt) e^{i\theta(\xi, t)}. \quad (17)$$

We then obtain from (14)

$$q(\xi, t) = \frac{\beta D |a(\xi, t)|^2}{MR (v_a^2 - v^2)}. \quad (18)$$

Substituting the right-hand side of (18) into (13) and (16), one obtains

$$i\hbar \frac{\partial a(\xi, t)}{\partial t} - \lambda a(\xi, t) + J \frac{\partial^2 a(\xi, t)}{\partial \xi^2} + G_r |a(\xi, t)|^2 a(\xi, t) = 0, \quad (19)$$

$$T + U = \frac{G_r}{2(v_a^2 - v^2)} \int_{-\infty}^{\infty} d\xi \left[\left(\int_{-\infty}^{\xi} d\eta \frac{\partial}{\partial t} |a(\eta, t)|^2 \right)^2 + v_a^2 |a(\xi, t)|^4 \right], \quad (20)$$

$$G_r \equiv \frac{\beta^2 D^2}{MR^2 (v_a^2 - v^2)} = \frac{G_0}{1 - V^2/V_a^2}, \quad (21)$$

$$G_0 \equiv \beta^2 D^2 / M V_a^2. \quad (22)$$

The exact solution of the nonlinear equation (19), normalized by

$$\int_{-\infty}^{\infty} |a(\xi, t)|^2 d\xi = 1$$

has the form

$$a_r(\xi, t) = \frac{\sqrt{\mu_r} \exp \left\{ i \left[\frac{\hbar v}{2J} (\xi - \xi_0) - \Omega_r t \right] \right\}}{\cosh [\mu_r (\xi - \xi_0 - vt)]}, \quad (23)$$

where ξ_0 is an arbitrary constant due to the translational invariance of (13) and (14). The other quantities contained in (23) are defined by

$$\mu_r = G_r / 4J, \quad (24)$$

$$\hbar \Omega_r = \lambda + (\hbar^2 v^2 - \frac{1}{4} G_r^2) / 4J. \quad (25)$$

When (23) is taken into account the expression (20) takes the form

$$T + U = \frac{G_r^2 (v^2 + v_a^2)}{24J (v_a^2 - v^2)}.$$

Substituting this into (15), we obtain the final expression for the energy of a soliton described by the function (23), when the velocity of its motion along the chain is fixed $V = vR$

$$\hbar \Omega_r = \hbar \Omega_0 + \frac{\hbar^2 V^2}{4JR^2} + \frac{G_0^2 (Z^2 + \frac{3}{2} Z^4 - \frac{1}{2} Z^6)}{24J (1 - Z^2)^3}, \quad (26)$$

where

$$Z \equiv V/V_a, \quad \hbar \Omega_0 = \varepsilon - D - 2J - G_0^2 / 48J \quad (27)$$

is the internal energy of the soliton.

It follows from the expression (26) that when the soliton velocity approaches the longitudinal sound velocity V_a , the soliton energy tends to infinity. At all finite values of the soliton energy its velocity is therefore less than V_a . The soliton energy thus cannot be transformed into the energy of the longitudinal waves.

At small velocities of the soliton ($V \ll V_a$) the soliton energy (26) takes the form

$$\hbar \Omega_r = \hbar \Omega_0 + \frac{\hbar^2 V^2}{2R^2 J} + \frac{G_0^2 V^2}{24 V_a^2 J} \left[1 - \frac{9}{2} (V/V_a)^2 + \dots \right].$$

In the same approximation the parameter

$$\mu_v \approx \frac{G_0}{4J} [1 + (V/V_a)^2] \approx \frac{G_0}{4J} = \mu_0 \quad (28)$$

depends but very weakly upon the soliton velocity.

Substituting (23) into (18) we find the distribution of the changes in the relative distances between the molecules

$$\varrho(\xi, t) = \frac{\sqrt{G_0^3/w}}{8J(1 - V^2/V_a^2)^2} \operatorname{sech} [\mu_v(\xi - \xi_0 - vt)] . \quad (29)$$

According to (29), the maximal deviation

$$\varrho_{\max} = \frac{\sqrt{G_0^3/w}}{8J(1 - V^2/V_a^2)^2} \quad (30)$$

obtained at $\xi = \xi_0 + vt$ moves along the chain with the velocity $V = vR$. The deformation moving along the chain is accompanied by the internal excitation. The region of the chain embraced by the internal excitation and the deformation is defined by

$$R \Delta\xi \approx \frac{\pi R}{\mu_v} = 4\pi J R / G_v . \quad (31)$$

The applicability condition of the continuum approximation is associated with the validity of the inequality $\Delta\xi = 4\pi J / G_v \gg 1$.

3. The Stability of Solitons

A remarkable property of solitons is that once the velocity is fixed the distribution of excitation and deformation described by the functions (23) and (29) remains unchanged during the soliton motion. At $V \ll V_a$ we have $G_v \approx G_0$, so that the size and shape of the soliton are both independent of its velocity, i.e. the soliton slows down without changing its shape.

In order to study the stability of solitons we consider the expression (10) at zero velocity ($v = 0$). In this case $a(\xi, t)$ represents a state with a simple time dependence, i.e.

$$a(\xi, t) = \Phi(\xi) \exp(-i\Omega_0 t) . \quad (32)$$

Using the continuum approximation in (10) and taking (32) into account, we obtain

$$F_0\{\Phi\} = \varepsilon - D - 2J - \int_{-\infty}^{\infty} \Phi(\xi) \left[J \frac{\partial^2 \Phi(\xi)}{\partial \xi^2} + \frac{1}{2} G_0 \Phi^3(\xi) \right] d\xi . \quad (33)$$

The expression (33) can be regarded as a functional of $\Phi(\xi)$; its absolute minimum defines the soliton energy at zero velocity, i.e.

$$(\hbar\Omega)_{\min} = \min F_0\{\Phi\} . \quad (33a)$$

If $\Phi(\xi)$ is taken to be a normalized function depending on a single parameter μ

$$\Phi(\xi) = \sqrt{\mu/2} \operatorname{sech} [\mu(\xi - \xi_0)] , \quad (34)$$

we obtain

$$F_0(\mu) = \varepsilon - D - 2J + \frac{1}{3} [J\mu^2 - \frac{1}{2} G_0\mu]. \quad (35)$$

The minimum (35) corresponds to the value $\mu_0 = G_0/4J$, and the energy

$$(\hbar\Omega)_{\min} = \varepsilon - D - 2J - G_0^2/48J \quad (36)$$

is exactly the same as that given by (27). The functions different from (34), or the values $\mu \neq \mu_0$ correspond to higher energies. In particular, the value of the functional (35) at $\mu = 0$ corresponds to nonlocalized excitations — the excitons. Their energy (at zero velocity) is equal to $\varepsilon - D - 2J$ thus exceeding the soliton energy by the value $G_0^2/48J$. These excitations are unstable in one-dimensional molecular systems.

If the electric dipole moment d is directed perpendicular to the chain, the parameter J which characterizes the resonance interaction will have a negative value ($\approx -d^2/R^3$). In this case the functional (35) has no minimum, the continuum approximation is not valid and there are no stable excitations such as the soliton. The excitation tends to be completely localized on a single molecule.

In conclusion we investigate the stability of the solitons for external disturbances. As a disturbance we consider a sound wave

$$A \cos \omega (t - \xi/v_a) \quad (37)$$

assuming the frequency ω to be low enough so that the sound wavelength exceeds considerably the size of an exciton,

$$\frac{2\pi v_a R}{\omega} \gg R \Delta\xi = 4\pi J R / G_0. \quad (38)$$

As mentioned above, at finite values of the soliton energy its velocity is much less than that of longitudinal sound waves. Therefore when investigating the soliton-longitudinal sound wave interaction the solitons may be regarded as unmoving.

To take into account the role of the exciton-sound wave interaction we investigate the solutions of (13) and (14) using

$$\varrho(\xi, t) = \varrho_0(\xi) + A \cos \omega(t - \xi/v_a), \quad (39)$$

Here

$$\varrho_0(\xi) = \frac{G_0}{8J} \sqrt{\frac{G_0}{w}} \operatorname{sech}^2(\mu_0 \xi), \quad \mu_0 = G_0/4J$$

is the solution for a free soliton at rest. In this case (14) is exactly satisfied by the function

$$a(\xi, t) = \Phi(\xi) \exp[-i\theta(t)], \quad (40)$$

if

$$\Phi(\xi) = \sqrt{\mu_0/2} \operatorname{sech} \mu_0 \xi. \quad (40a)$$

At $|\xi| > \Delta\xi$ equation (13) is satisfied approximately, the function (40) and its derivatives tending rapidly to zero. In the region $|\xi| \leq \Delta\xi$, under the condition (38) the sound wave has the same phase. Equation (13) thus reduces to

$$\frac{\partial \theta(t)}{\partial t} + \frac{\beta D A}{\hbar R} \cos \omega t - \Omega_0 = 0. \quad (41)$$

The soliton energy is defined by the integral

$$E = i\hbar \int a^*(\xi, t) \frac{\partial a(\xi, t)}{\partial t} d\xi.$$

Therefore, when the expressions (40) and (41) are taken into account we have

$$E = \hbar \frac{\partial \theta(t)}{\partial t} = \hbar \Omega_0 - \frac{\beta D A}{R} \cos \omega t. \quad (42)$$

The average value of soliton energy (when it is averaged over the period of a sound wave) is thus equal to the energy $\hbar \Omega_0$ of a free soliton. According to (40), the sound wave does not change the shape of a soliton, defined by the function $\Phi^2(\xi) = (\mu_0/2) \operatorname{sech}^2(\mu_0 \xi)$.

Thus the excitations of a molecular chain in the form of solitons have great stability in retaining the shape, magnitude, and the size of the region occupied by the excitation. This fact has been used by one of the authors [2] to account for the great efficiency of the transfer of energy released in the hydrolysis of ATP molecules in living organisms along the α -helical protein molecules and to elucidate the contraction mechanism of transversely striated animal muscles at the molecular level [4].

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(Received March 24, 1976)